Smart Infrastructure

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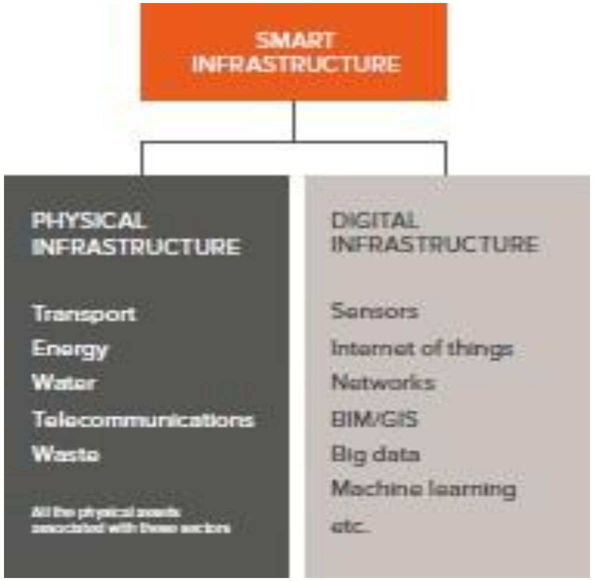
- Smart Infrastructure is the result of combining physical infrastructure with digital infrastructure, providing improved information to enable better decision making, faster and cheaper.



What is Smart Infrastructure

At the core, being “Smart” is understanding how to plan, integrate and operate technologies holistically

Financial, social and political pressures drive infrastructure owners to improve end user service in more cost- and resource-efficient ways

Added infrastructure intelligence enables increased reliability, efficiency and security while enhancing the end user experience and creating a sustainable future

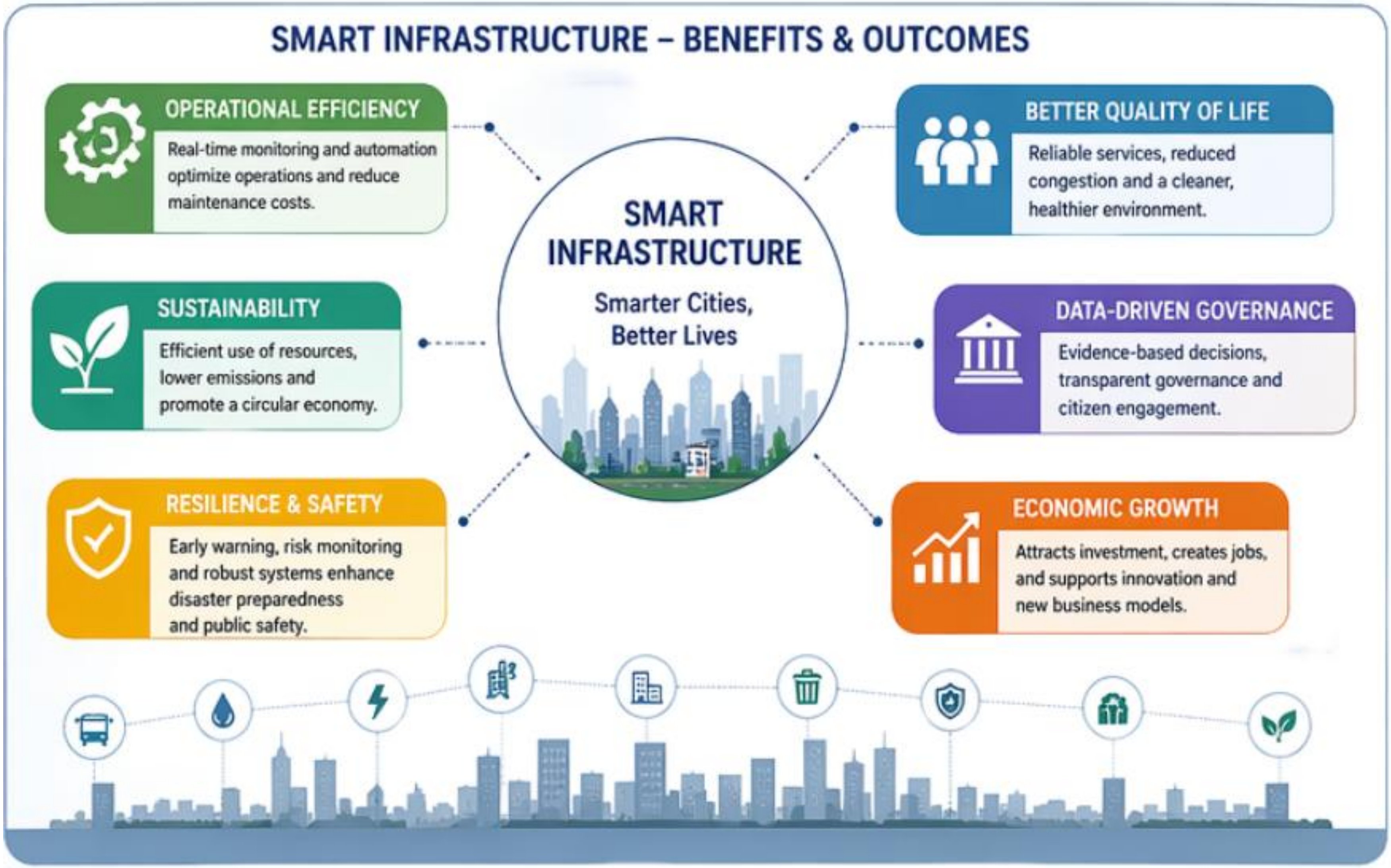
Why Smart Infrastructure

1.Smart Infrastructure will allow owners and operators to get more out of what they already have – increasing capacity, efficiency, reliability and resilience.

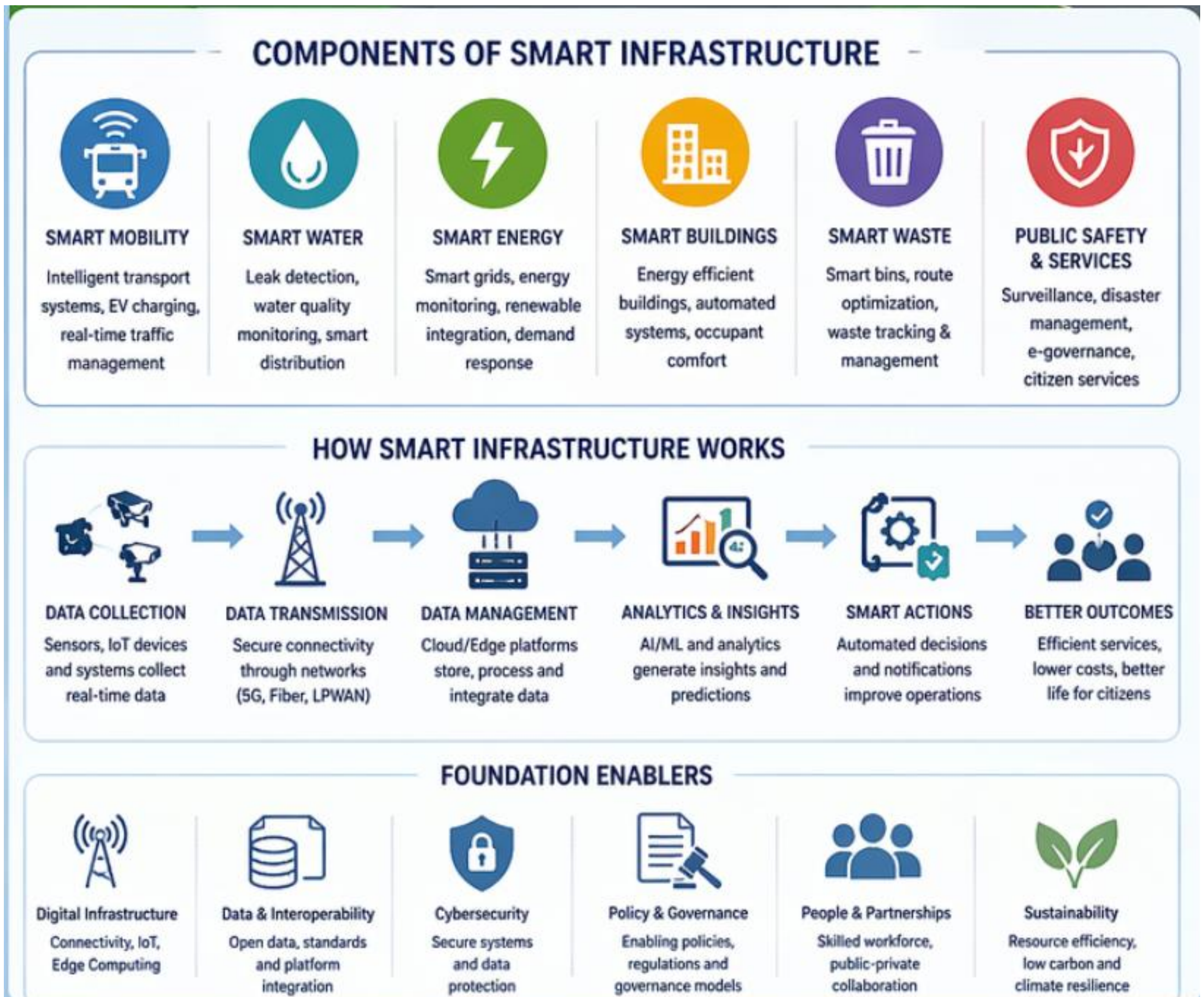
2. Getting more from existing assets will enable owners and operators to enhance service provision despite constrained finance, growing resource scarcity and, in mature economies, short supply of green field space.

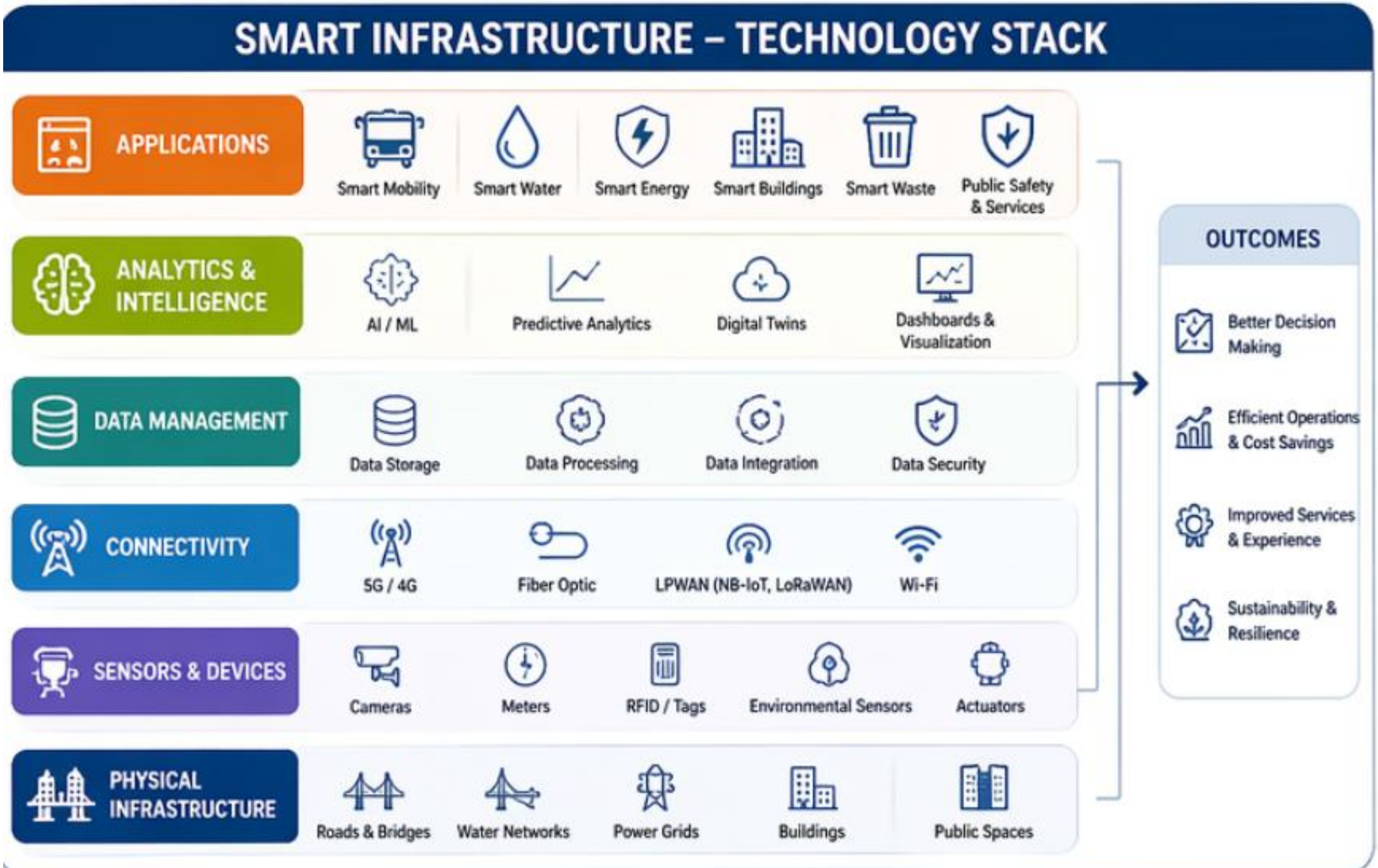
3. Better understanding of the performance of our infrastructure will allow new infrastructure to be designed and delivered more efficiently, and to provide better whole-life value.

4. Owners will be able to use technology to gather data from their infrastructure in use – monitoring condition so maintenance is targeted at the right time; and engaging directly with customers to manage demand and reduce the need for new construction.



SMART INFRASTRUCTURE – BENEFITS & OUTCOMES
OPERATIONAL EFFICIENCY
Real-time monitoring and automation optimize operations and reduce maintenance costs.
SUSTAINABILITY
Efficient use of resources, lower emissions and promote a circular economy.
RESILIENCE & SAFETY
Early warning, risk monitoring and robust systems enhance disaster preparedness and public safety.
SMART INFRASTRUCTURE
Smarter Cities, Better Lives
BETTER QUALITY OF LIFE
Reliable services, reduced congestion and a cleaner, healthier environment.
DATA-DRIVEN GOVERNANCE
Evidence-based decisions, transparent governance and citizen engagement.
ECONOMIC GROWTH
Attracts investment, creates jobs, and supports innovation and new business models.


COMPONENTS OF SMART INFRASTRUCTURE
SMART MOBILITY
Intelligent transport systems, EV charging, real-time traffic management
SMART WATER
Leak detection, water quality monitoring, smart distribution
SMART ENERGY
Smart grids, energy monitoring, renewable integration, demand response
SMART BUILDINGS
Energy efficient buildings, automated systems, occupant comfort
SMART WASTE
Smart bins, route optimization, waste tracking & management
PUBLIC SAFETY & SERVICES
Surveillance, disaster management, e-governance, citizen services
HOW SMART INFRASTRUCTURE WORKS
DATA COLLECTION
Sensors, IoT devices and systems collect real-time data
DATA TRANSMISSION
Secure connectivity through networks (5G, Fiber, LPWAN)
DATA MANAGEMENT
Cloud/Edge platforms store, process and integrate data
ANALYTICS & INSIGHTS
AI/ML and analytics generate insights and predictions
SMART ACTIONS
Automated decisions and notifications improve operations
BETTER OUTCOMES
Efficient services, lower costs, better life for citizens
FOUNDATION ENABLERS
Digital Infrastructure
Connectivity, IoT, Edge Computing
Data & Interoperability
Open data, standards and platform integration
Cybersecurity
Secure systems and data protection
Policy & Governance
Enabling policies, regulations and governance models
People & Partnerships
Skilled workforce, public-private collaboration
Sustainability
Resource efficiency, low carbon and climate resilience


SMART INFRASTRUCTURE – TECHNOLOGY STACK
APPLICATIONS
Smart Mobility
Smart Water
Smart Energy
Smart Buildings
Smart Waste
Public Safety & Services
ANALYTICS & INTELLIGENCE
AI / ML
Predictive Analytics
Digital Twins
Dashboards & Visualization
DATA MANAGEMENT
Data Storage
Data Processing
Data Integration
Data Security
CONNECTIVITY
5G / 4G
Fiber Optic
LPWAN (NB-IoT, LoRaWAN)
Wi-Fi
SENSORS & DEVICES
Cameras
Meters
RFID / Tags
Environmental Sensors
Actuators
PHYSICAL INFRASTRUCTURE
Roads & Bridges
Water Networks
Power Grids
Buildings
Public Spaces
OUTCOMES
Better Decision Making
Efficient Operations & Cost Savings
Improved Services & Experience
Sustainability & Resilience

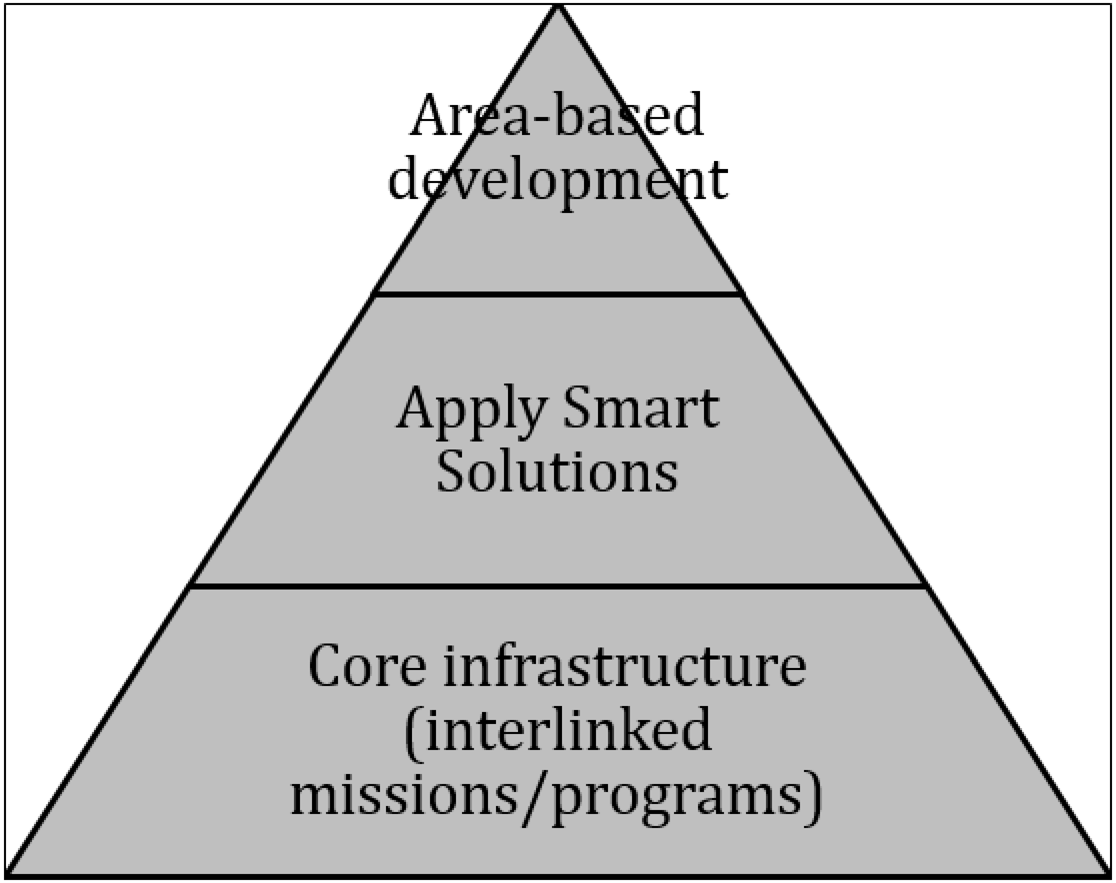
IN PURSUIT OF TRANSFORMATION: 100 SMART CITY MISSION

- Government of India converted Existing Cities in to Smart Cities
- Existing Cities will be required to augment their basic infrastructure, this would encourage Urban Migration
- A better approach would have been to develop new areas of economic growth & employment generation along the energy corridor.
- The 100 Smart Cities mission generated huge public enthusiasm. However it became a check box oriented approach for competition among the various cities. The focus shifted on individual ranking of cities.
- **The development of urban habitats with a robust sustainable environmental resilience was missing from the agenda….**

Smart City

- **Objective:** To promote cities that provide core infrastructure and give a decent quality of life to its citizens, a clean and sustainable environment and application of IT enabled smart solutions.
- **Strategy:** Apply smart solutions to improve governance and public services and make the city smart through area-based development.

Holistic Development
three **layers** of smartness



Holistic Development three **layers** of smartness

<u>First layer</u>: Provides citywide core infrastructure (basic services and physical infrastructure) to provide for the basic needs of the residents. This is the first layer of the smart city and some of the elements are given below.

- Adequate and clean water supply,
- Assured electricity supply,
- Sanitation facilities, including and solid waste management,
- Efficient urban mobility and public transport,
- Affordable housing, especially for the poor,
- Robust IT connectivity and digitalization,
- Good governance, especially. e-Governance and citizen participation,
- Sustainable environment,
- Safety and security of citizens, particularly women, children and the elderly, and
- Health and education.

Second layer: Consists of application of smart solutions to infrastructure and services to improve them1.

Application of smart solutions to existing infrastructure and services is expected to,

(1) extract more output from the existing assets and resources,

(2) use less resources to deliver the same output and,

(3) improve governance and delivery of public services.

Third layer: This consists of development of all elements in compact parts of the city, called '**Areas**' (also called area-based developments and used interchangeably). Such Areas act as a **Lighthouse** for other Areas in the City to emulate and develop over time

- **Retrofitting** will completely develop an existing built area of more than 500acres so as to achieve Smart City objectives, along with other objectives of making the existing area more efficient and livable.
- **Redevelopment** will replace the existing built environment in an area greater than 50 acres and enable co-creation of a new layout, especially enhanced infrastructure, mixed land use and increased density.
- **Greenfield** developments will completely develop a previously vacant area, of more than 250 acres, using innovative planning, plan financing and plan implementation tools with provision for affordable housing, especially for the poor

SMART CITY: FUNDAMENTALS

SMART ENVIRONMENT

SMART HEALTH

SMART GOVERNANCE

Smart Cities Selection Process

Letter to all state governments to shortlist potential Smart Cities based on Stage-I criteria according to number of Smart Cities distributed across states /UTs by the MoUD. This is the first stage of Intra-State competition.

On the basis of response from States/UTs, the list of potential 100 Smart Cities is announced. The second stage of All India competition begins.

Each potential Smart City prepares its proposal assisted by a consultant (from a panel prepared by MoUD) and a hand-holding External Agency (various offers received such as World Bank, ADB, GEF, USTDA, JICA, DFID, AFD,KfW, UN Habitat)

By stipulated date Stage 2 proposals submitted. Evaluation by a panel of experts.

Selected cities declared – Round 1 Smart Cities

Selected cities set up SPV and start implementation of their SCP. Preparation of DPRs, tenders etc.

Other cities prepare to improve their proposal for next round of the Challenge

Scoring criteria

Area-based development 55%

7%
'Smartness' of proposal

3%
Process followed

5%
Citizen engagement

25%
Implementation framework, including feasibility and cost-effectiveness

15%
Results orientation

City Level Criteria 30%

5%
Vision and goals

10%
Citizen engagement

10%
Strategic plan

5%
Baseline, Key Performance Indicators (KPIs), self-assessment and potential for improvement

Pan-city solution 15%

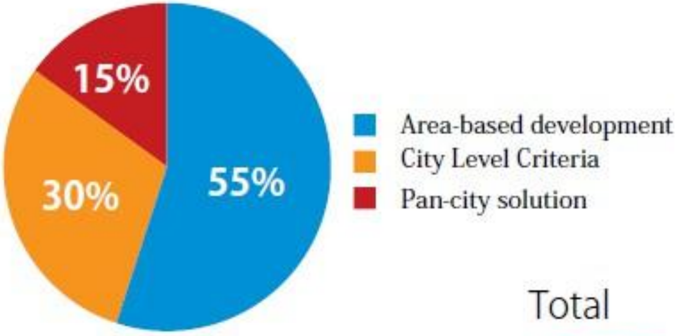
3%
"Smartness' of solution

1%
Process followed

1 %
Citizen engagement

5 %
Implementation framework, including feasibility and cost-effectiveness

5 %
Results orientation



Total

100

Dovetailing of interlocked programs around the smart city mission

Eight Critical Pillar of Smart Cities in India

1. Smart Governance: Investments of about US$1.2 trillion will be required over the next 20 years across areas such as transportation, energy and public security to build smart cities in India. Highlights:

- US$1.2 billion allocated for smart cities and FDI norms relaxed
- US$83 million allocated for Digital India Initiative
- PPP Model to be used to upgrade infrastructure in 500 urban areas
- Smart City projects to create 10-15% rise in employment
- Ministry of Urban Development has plans to develop 2 smart cities in each of India's 29 states
- Delhi Mumbai Industrial Corridor Development Corporation Ltd (DMICDC) plans seven "smart cities" along the 1,500 km industrial corridor across six states with a total investment of US$100 billion

2. Smart Energy: Three crucial dimensions of smart energy systems are: Smart Grid

- Electrification of all households with power available for at least 8 hours per day by 2017
- Implementation of 8 smart grid pilot projects in India with an investment of US$10 million

Energy Storage

- Addition of 88,000 MW of power generation capacity in the twelfth five year plan (2012-17)
- India needs to add at least 250-400 GW of new power generation capacity by 2030
- The Power Grid Corporation of India has planned to invest US$26 billion in the next five years
- Smart Meters
- India to install 130 million smart meters by 2021

3. Smart Environment: Three crucial dimensions of ensuring sustainable development are: Renewable Energy

- Ministry of New and Renewable Energy has plans to add capacity of 30,000 MW in the 12th Five Year Plan (2012-17)
- Water and Waste Water Management
- The Indian Ministry of Water Resources plans to invest US$50 billion in the water sector in the coming years
- The Yamuna Action Plan Phase III project for Delhi is approved at an estimated cost of US$276 million
- Sanitation
- The Government of India and the World Bank have signed a US$500 million credit for the Rural Water Supply and Sanitation (RWSS) project in the Indian states of Assam, Bihar, Jharkhand and Uttar Pradesh

4. Smart Transportation: The Government of India has set ambitious targets of developing public transportation system to support the ever growing urban populace. Green Transport

- The Government of India has approved a US$4.13 billion plan to spur electric and hybrid vehicle production by setting an ambitious target of 6 million vehicles by 2020
- Electric vehicle charging stations in all urban areas and along all state and national highways by 2027
- Metro: Ministry of Urban Development plans to invest more than US$20 billion on the metro rail projects in coming years
- High Speed Rail: The proposed 534 km Mumbai-Ahmedabad high speed rail project will have an investment of around US$10.5 billion
- Monorail: India’s first monorail project at Mumbai will cost around US$500 million, of which US$183 million has been spent on phase I

5. Smart IT & Communications: Information and Communications Technology

- Cloud computing will evolve into a US$4.5 billion market in India by 2016
- Broadband connections to 175 million users by 2017

Security and Surveillance

- Under the flagship “Safe City” project, the Union Ministry proposes US$333 million to make seven big cities (Delhi, Mumbai, Kolkata, Chennai, Ahmedabad, Bangalore and Hyderabad) to focus on technological advancement rather than manpower Disaster Management
- The Government of India and World Bank signed US$236 million agreement for reducing disaster risks in coastal villages of Tamil Nadu and Puducherry

6. Smart Buildings:

- India is expected to emerge as the world’s 3rd largest construction market by 2020, by adding 11.5 million homes every year
- The Intelligent Building Management Systems market is around US$621 million and is expected to reach US$1,891 million by 2016
- Smart Buildings will save up to 30% of water usage, 40% of energy usage and reduction of building maintenance costs by 10 to 30%

7. Smart Health Hospitals

- Health budget up by 27% in FY 2014-15 to US$5.26 billion, with special focus on improving affordable healthcare for all
- To establish six new AIIMS like institutes and 12 government medical colleges in the country
- Accessible, affordable and effective healthcare system for 1.2+ billion citizens

Insurance

- FDI limit in the insurance sector increased to 49% from 26%
- Insurance industry has potential to reach US$1 trillion by 2020

Medical Devices

- Indian medical devices market to reach US$11 billion by 2023
- 100% FDI allowed in the medical devices sector under the automatic route

Wellness

- Indian wellness industry is expected to reach around US$16.65 billion by 2015

8. Smart Education: The Government of India has allocated US$13.95 billion in the Union Budget 2014-15 for the education sector, up by 12.3% from the previous year.

- Budget has allocated US$78.5 million to set-up five new IITs and five new IIMs
- The Ministry of Human Resource Development plans to have 1,000 private universities for producing trained manpower to meet services and industry requirements
- 100% FDI allowed in the education sector
- India’s online education market size expected to be US$40 billion by 2017


SMART INFRASTRUCTURE
Building Intelligent, Connected and Sustainable Cities
Smart infrastructure uses data, technology and innovation to plan, build, operate and maintain assets efficiently for a better quality of life.
SMART MOBILITY
SMART WATER
SMART ENERGY
SMART BUILDINGS
SMART WASTE
PUBLIC SAFETY & SECURITY
SMART CITIZEN SERVICES
WHAT IS SMART INFRASTRUCTURE?
Smart infrastructure integrates physical assets with digital technologies and real-time data to improve efficiency, resilience, sustainability and service delivery.
HOW IT WORKS
SENSORS & DEVICES
Collect real-time data
DATA TRANSMISSION
Data is securely sent to platforms
DATA ANALYTICS
Insights generated using AI/ML
SMART DECISIONS
Better planning, operations & services
KEY TECHNOLOGIES
Internet of Things (IoT)
Artificial Intelligence (AI) & Machine Learning (ML)
Big Data & Analytics
Cloud Computing
GIS & Digital Twins
5G/Advanced Connectivity
Blockchain (for security & transparency)
KEY BENEFITS
EFFICIENCY
Optimized operations and resource use reduce costs.
SUSTAINABILITY
Lower carbon footprint, better energy and water management.
RESILIENCE
Real-time monitoring helps in disaster preparedness and faster response.
QUALITY OF LIFE
Better services, safer communities and enhanced livability.
ECONOMIC GROWTH
Attracts investment, creates jobs and supports innovation.
EXAMPLES OF APPLICATIONS
Adaptive traffic signals & smart parking
Leak detection in water supply networks
Predictive maintenance of assets
Smart street lighting
Flood monitoring & early warning systems
Integrated Command & Control Centers
ENABLERS FOR SMART INFRASTRUCTURE
Strong Policy & Governance
Digital Infrastructure & Connectivity
Standards & Interoperability
Data Security & Privacy
Skilled Workforce & Capacity Building
Public-Private Partnerships
Citizen Engagement
SMART INFRASTRUCTURE LIFECYCLE
PLAN
Data-driven planning & feasibility
DESIGN
Use digital tools, BIM & simulation
BUILD
Smart construction & progress monitoring
OPERATE
Real-time monitoring & efficient operations
MAINTAIN
Predictive maintenance & asset management
RENEW/UPGRADE
Data-driven retrofits & continuous improvement
SMART INFRASTRUCTURE FOR A SMARTER TOMORROW
Connected Cities
Efficient Infrastructure
Sustainable Environment
Resilient Communities
Better Tomorrow
Integrating people, technology and infrastructure for a better, smarter future.



Thank you !